

UniVerse

A Campus Companion

Teacher Features

The teacher Home dashboard and Manage Classes - cancel a class, post a notice, and undo, with live student alerts.

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

Owner: Robi (student/teacher features)

Key files: features/teacher/* (+ cancellations table, migration 007)

1. Two teacher screens

- **Teacher Dashboard** - same idea as the student home, but built from ``fetchForTeacher``: greeting, live/next class hero, today's classes, stats.
- **Manage Classes** - the teacher's action centre: pick a day, tap a class, then Cancel it / Post an update / Undo. This is where teachers WRITE to the backend.

2. Cancelling a class - the full flow

A cancellation targets ONE specific occurrence (a single date), not the whole weekly slot. ``occurrenceDate(day)`` computes the next calendar date ($\geq$ today) whose weekday matches the class's day.

- 1 Teacher taps a class -> chooses 'Cancel' -> types a reason.
- 2 ``TeacherService.cancelClass()`` INSERTs a row into **cancellations** (routine_id + class_date + reason + cohort info + cancelled_by).
- 3 In the SAME call it fires a broadcast: ``createBroadcast(classCancel, target = that batch+section)``.
- 4 Inserting that notification row triggers BOTH the in-app Realtime alert AND the OS push (via the send-push Edge Function).
- 5 The teacher's own card badges the class as CANCELLED (the key is added to a local cancelled set).

So one tap = a durable cancellation record + an instant alert to exactly the affected students. The student grid badge isn't wired yet - students get the alert - but the data model supports it.

3. Post a notice / Undo

- **Post update** - ``postNotice()`` sends a room-change / notice / test-reminder broadcast to the class's cohort (no cancellation row, just a notification).
- **Undo** - ``undoCancel()`` deletes the cancellation row (scoped to ``cancelled_by = me`` by RLS). The already-sent alert stays in students' history - you can't recall a sent message, and that's honest behaviour.

4. The cancellations table (migration 007)

One dated row per cancelled occurrence: ``routine_id, class_date, reason, batch, section, subject, day, time_start, cancelled_by``, with a UNIQUE constraint on ``(routine_id, class_date)`` so the same class can't be cancelled twice for the same date.

RLS rule	who
Read	Everyone (so teacher view + future student view can badge CANCELLED).
Insert / Delete	Only the teacher/admin who owns it (<code>cancelled_by = auth.uid()</code>).
Realtime	Enabled, so changes can stream to clients.

5. Likely defense questions

question	answer
Does cancelling delete the routine row?	No - the weekly routine is permanent. A cancellation is a separate dated exception row.

**How do students find out
instantly?**

The same insert that records the cancellation also inserts a notification -> Realtime + push fan out to the cohort.

Can a teacher cancel	No - RLS ties insert/delete to cancelled_by = auth.uid() plus the teacher/admin role.
Why target batch+section?	createBroadcast sets target_batch/target_section so only the affected cohort is alerted, not the whole campus.